Yuqing REN

Algorithms and Implementations for B5G/6G Baseband
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EDUCATION EXPREIENCES

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland *2021.05–2025.08*
Ph.D., Electrical Engineering
Advisor: Prof. Andreas Burg
Thesis Title: Efficient Algorithms and VLSI Implementations of FEC Decoders for B5G/6G
Southeast University, Nanjing, China *2018.09–2021.05*
M.S., Electronics and Communication Engineering
Advisor: Prof. Xiaohu You
Nanjing University of Science and Technology, Nanjing, China *2014.09 –2018.06*
B.S., Electronic and Information Engineering

WORKING EXPERIENCES

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Swiss *2025.08–now*
Postdoc researcher. Collaborating Advisor: Prof. Andreas Burg

HONORS AND AWARDS

2025	EDEE PhD Thesis Award, EPFL	Top 1/college
2025	IEEE CASS Pre-Doctoral Grants, *CAS Society*	Top 10/world
2024	Selected as highlighted paper in IEEE Trans. CAS-I, *CAS Society*	
2024	Chinese Government Award for Outstanding Self-Financed Students Abroad	Top 650/world
2022	Outstanding Master's Thesis of Jiangsu Province	Top 150/province
2015	National Scholarship for Undergraduate Student	1/70

PROFESSIONAL ACTIVITIES

Journal Reviewer

- IEEE Transactions on Signal Processing (TSP)
- IEEE Transactions on Communications (TCOM)
- IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)
- IEEE Solid-State Circuits Letters (SSCL)
- IEEE Signal Processing Letters (SPL), IEEE Communications Letters (CMML), IEEE TCAS-II

Conference Reviewer

- ISCAS 2025/2024/2023/2020, Globecom 2025, ISTC 2025, SiPS 2024/2018, GlobalSiP 2019, DSP 2018

TEACHING EXPERIENCES

EE334 *Digital System Design*	TA	*2021/2022/2023/2024*
EE342 *Systèmes de Télécommunications*	TA	*2024*
EE543 *Advanced Wireless Receivers*	TA	*2023*

Supervised a total of 3 undergraduate students and 5 master's students to complete 6 semester projects and guided 1 master's thesis (Title: Standard-Cell Memory Compiler for FEC Decoder)

PUBLICATIONS

Journal Papers

1. **Y. Ren**, L. Zhang, Y. Shen, W. Song, E. Boutillon, A. Balatsoukas-Stimming, A. Burg, "Edge-spreading Raptor-like LDPC codes for 6G wireless systems," accepted by ***IEEE Transactions on Communications (TCOM)***, 2025.
2. **Y. Ren**, H. Harb, Y. Shen, A. Balatsoukas-Stimming, A. Burg, "A Generalized Adjusted Min-Sum Decoder for 5G LDPC Codes: Algorithm and Implementation," ***IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)***, vol. 71, no. 6, Jun. 2024. **Highlighted in June Issue (LinkedIn)**
3. **Y. Ren**, L. Zhang, L. D. Blanc, Y. Shen, X. Li, A. Balatsoukas-Stimming, C. Zhang, A. Burg, "A Node-Based Polar List Decoder With Frame Interleaving and Ensemble Decoding Support," accepted by ***IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)***, 2024. **Special Issue invited by IEEE ISCAS**
4. **Y. Ren**, Y. Shen, L. Zhang, A. T. Kristensen, A. Balatsoukas-Stimming, E. Boutillon, A. Burg, C. Zhang, "High-Throughput and Flexible Belief Propagation List Decoder for Polar Codes," ***IEEE Transactions on Signal Processing (TSP)***, vol. 72, Feb. 2024.
5. **Y. Ren**, A. T. Kristensen, Y. Shen, A. Balatsoukas-Stimming, C. Zhang, A. Burg, "A Sequence Repetition Node-Based Successive Cancellation List Decoder for 5G Polar Codes: Algorithm and Implementation," ***IEEE Transactions on Signal Processing (TSP)***, vol. 70, Oct. 2022.
6. **Y. Ren**, Y. Shen, Z. Zhang, X. You, and C. Zhang, "Efficient Belief Propagation Polar Decoder With Loop Simplification Based Factor Graphs," ***IEEE Transactions on Vehicular Technology (TVT)***, vol. 69, May 2020.
7. Y. Sun, J. Chen, **Y. Ren**, H. Ji, Y. Huang, X. You, C. Zhang, "GenPolar: Generative AI-Aided Complexity Reduction for Polar SCL Decoding," accepted by *IEEE Journal of Emerging and Selected Topics in Circuits and Systems (JETCAS)*, 2025.
8. Y. Shen, Z. Li, **Y. Ren**, E. Boutillon, A. Balatsoukas-Stimming, X. You, C. Zhang, A. Burg, "Towards Universal Belief Propagation Decoding for Short Binary Block Codes," accepted by ***IEEE Journal of Selected Areas in Communications (JSAC)***, 2024.
9. Z. Li, Y. Shen, **Y. Ren**, Y. Huang, X. You, C. Zhang, "Belief Propagation Decoding for Short-Length Codes Based on Sparse Tanner Graph," ***IEEE Communications Letters (CMML)***, vol. 28, no.. 5, May 2024.
10. M. Hashemipour-Nazari, **Y. Ren**, K. Goossens and A. Balatsoukas-Stimming, "Pipelined Architecture for Soft-Decision Iterative Projection Aggregation Decoding for RM Codes," ***IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)***, vol. 70, no. 12, pp. 5468-5481, Dec. 2023.
11. Y. Shen, W. Song, H. Ji, **Y. Ren**, C. Ji, X. You, and C. Zhang, "Improved Belief Propagation Polar Decoders with Bit-Flipping Algorithms," ***IEEE Transactions on Communications (TCOM)***, vol. 68, no. 11, November, 2020.
12. Y. Shen, W. Song, **Y. Ren**, H. Ji, X. You, and C. Zhang, "Enhanced Belief Propagation Decoder for Polar Codes with Bit-Flipping," ***IEEE Transactions on Circuits and Systems II-Express Briefs (TCAS-II)***, vol. 67, May 2020.

Conference papers

1. **(Oral) Y. Ren**, L. Zhang, E. Boutillon, A. Balatsoukas-Stimming, A. Burg, "A Universal LDPC Matrix for Short-Length Codes and Long-Length Codes," accepted by ***Proc. IEEE Globe. Comm. Conf. (Globecom)***, Dec., 2025.
2. **(Oral) Y. Ren**, Y. Shen, W. Song, A. Balatsoukas-Stimming, E. Boutillon, A. Burg, "Towards 6G: Configurable high-throughput decoder implementation for SC-LDPC codes," in ***Proc. IEEE Asilomar Conf. Sign. Sysm. Comp (Asilomar)***, Oct., USA, 2024.
3. **(Oral) Y. Ren**, Y. Shen, E. Boutillon, A. Balatsoukas-Stimming, A. Burg, "Design of Concatenated Product Codes with Spatially-Coupled LDPC Codes," in ***Proc. IEEE Int. Symp. Wireless Commu. Syst. (ISWCS)***, Brazil, 2024.
4. **(Oral) Y. Ren**, W. Xu, X. You, and C. Zhang, "Efficient Belief Propagation List Decoding of Polar Codes," in ***Proc. IEEE Int. Conf. on ASIC (ASICON)***, Nov, China, 2019.
5. **(Poster) Y. Ren**, F. Shu, L. Li, Z. Zhang, X. You, and C. Zhang, "A Novel D-Metric for Blind Detection of Polar Codes," in ***Proc. IEEE Int. Workshop Sign. Process. (SiPS)***, Oct. South Africa, 2018.
6. **(Oral)** Y. Shen, W. Song, L. D. Blanc, **Y. Ren**, A. Balatsoukas-Stimming, A. Alvarado, A. Burg, "Iterative Logistic Weight Based Chase Decoder for Open Forward Error Correction," in *Proc. Optical Fiber Comm. Conf. (OFC)*, USA, 2025.

7. **(Oral)** L. D. Blanc, V. Herrmann, **Y. Ren**, C. Muller, A. T. Kristensen, A. Levisse, Y. Shen, A. Burg, "A GRANDAB Decoder with 8.48 Gbps Worst-Case Throughput in 65nm CMOS," in ***Proc. IEEE Euro. Solid-State Elec. Research Conf. (ESSERC)***, Belgium, 2024.
8. **(Oral)** L. Zhang, **Y. Ren**, Y. Shen, W. Zhou, A. Balatsoukas-Stimming, A. Burg, "A low-latency and high-performance SCL decoder with frame-interleaving," in ***Proc. IEEE Int. Symp. Circuit Syst. (ISCAS)***, May, Singapore, 2024.
9. **(Oral)** Y. Shen, **Y. Ren**, and A. Burg, "Iterative Ordered Statistics Decoding of Product Codes," in ***Proc. IEEE Int. Symp. Topics Coding (ISTC), September***, *France*, 2023.
10. **(Poster)** Y. Shen, **Y. Ren**, A. T. Kristensen, X. You, C. Zhang, and A. Burg, "Improved Belief Propagation Decoding of Turbo Codes," in ***Proc. IEEE Int. Conf. Acoustics Speech Sign. Process. (ICASSP)***, *Greece*, 2023.
11. **(Oral)** Y. Shen, **Y. Ren**, A. T. Kristensen, A. Balatsoukas-Stimming, X. You, C. Zhang, and A. Burg "Fast Sequence Repetition Node-Based Successive Cancellation List Decoding for Polar Codes," in ***Proc. IEEE Int. Conf. Comm. (ICC)***, May, Korea, 2022.

INVITED TALKS

1. **From 5G Standard to Next-Generation: Channel Codes, Algorithms, and Implementations**
   Zhejiang University, China, Sep. 2024
   Soochow University, China, Dec. 2024

RESEARCH OVERVIEW

Research topic: Efficient Algorithms and VLSI Implementations for B5G/6G

- FEC decoders are one of the key bottlenecks determining the performance of B5G/6G digital baseband chips. With the slowdown of Moore's Law, **exploring novel designs and joint optimization of FEC algorithms and architectures** has become increasingly crucial. The goal is to achieve an optimal trade-off between error correction performance and hardware efficiency in digital baseband chips.
- Successful tape-outs at 16nm, 28nm, and 65nm process nodes.

Project 1: Advanced Decoding Algorithms and Hardware Architectures for B5G

- For the first time, a general node, Sequence Repetition (SR), was extended to the Successive Cancellation List (SCL) decoding algorithm for Polar codes. This work successfully implemented **the first list decoder based on generalized SR nodes (SR-List) in hardware**. Compared to state-of-the-art designs, the proposed SR-List decoder achieves nearly 2× throughput improvement. The 28nm ASIC achieves 2.94 Gbps throughput and 6.70 Gbps/mm² area efficiency with a list size of 8, providing a highly competitive solution to address the latency challenges of Polar codes in 5G-NR and B5G.
- To meet the stringent requirements of 5G-NR for LDPC decoding algorithms and VLSI implementation, this work **designs a high-performance, low-complexity 5G-NR LDPC decoder**. To bridge the performance gap between Belief Propagation (BP) decoding and its hardware approximations, a Generalized Adjusted Min-Sum (GA-MS) decoding algorithm was proposed. Under various 5G standard scenarios, GA-MS fixed-point performance closely approaches BP floating-point performance within a 0.1 dB gap. The 28nm ASIC LDPC decoder occupies a core area of 1.823 mm², operates at 895 MHz, and achieves a peak throughput of 24.42 Gbps with a maximum area efficiency of 13.40 Gbps/mm², outperforming other comparable works.
- The academic papers resulting from this project have been published in ***IEEE Transactions on Signal Processing*** and ***IEEE Transactions on Circuits and Systems I: Regular Papers*** **(refer to Journal papers [5] and [2] for details)**. Additionally, the relative works were featured in **keynote talks by Prof. Andreas Burg** at the *IEEE Global Communications Conference 2024 (link: https://globecom2024.ieee-globecom.org/workshop/ws-06-workshop-channel-coding-beyond-5g/program) and IEEE International Symposium on Topics in Coding 2023 (link: https://istc2023.org/keynote-presentations)*.

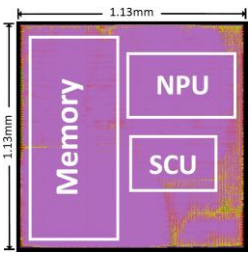
Project 2: Next-Generation Channel Coding and Tb/s Decoder Chip Towards 6G

- Next-generation channel coding faces extremely high requirements in terms of data rate, energy efficiency, and error rate, while also maintaining key properties of the 5G-NR standard codes, such as rate compatibility, making it a significant challenge. Due to outstanding capacity-approaching performance, Spatially Coupled LDPC (SC-LDPC) codes are considered a promising candidate for next-generation channel coding. This work proposes a novel SC-LDPC code for 6G, called Edge-Spreading Raptor-Like (ESRL) LDPC code. ESRL codes focus on constructing optimal coupling matrices and exhibit extremely high parallel processing capability. Compared to 5G-NR, this coding scheme demonstrates significant advantages in error rate performance, throughput, and hardware complexity, making it a promising solution for 6G wireless systems.
- Tb/s FEC decoders represent a critical bottleneck in 6G baseband chips, particularly for terahertz communications. This work presents **a B5G/6G dual-mode LDPC decoder chip, LIBRA**, **implemented in TSMC 16nm CMOS**. The ASIC occupies only 2.56 mm² of core area while achieving a peak throughput of over 100 Gbps.
- The core academic papers on algorithmic theory have been published in ***IEEE Transactions on Communications*** (**refer to Journal papers [1] for details**). The core academic papers on chip implementation (taped out on July/23/2025) are still under preparation, and **Prof. Andreas Burg has delivered an invited keynote lecture on this work** at *Huawei Future of Wireless 2024* (Stockholm, Sweden) (link: https://futureofwireless.eu/speakers-2024/).

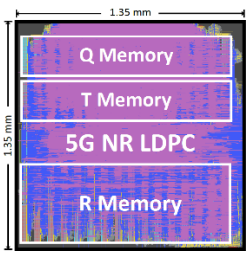
Project 3: Unified Coding and Decoding Architectures for Future Baseband Chips

- To address the coexistence of multiple standards and protocols in future B5G/6G systems, this work proposes a universal LDPC matrix design. By employing a flexible coupling mechanism, block codes and coupled codes are unified within a single construction framework, enabling seamless transition between short- and long-length codes. In long-code scenarios, the design inherits the excellent convergence properties and high-throughput advantages of coupled codes; in short-code scenarios, combined with ensemble list decoding, it achieves near–maximum-likelihood performance. This unified design supports both eMBB+ and uRLLC+ within the same underlying structure, providing a representative candidate for 6G.
- A unified decoder architecture based on ensemble list decoding is designed, which supports multiple classical algorithms and leverages a flexible permutation-graph generation mechanism to realize ensemble list decoding. In addition, a unified decoder chip based on noise-guessing has been implemented. Fabricated in 65nm CMOS technology, the chip achieves a guaranteed worst-case throughput of 8.48 Gbps, significantly outperforming existing counterparts, thereby demonstrating the feasibility and advantages of unified decoders in hardware.
- The core academic outcomes of this project have been published in ***IEEE Transactions on Signal Processing*** and ***IEEE Transactions on Vehicular Technology*** **(refer to Journal papers [4] and [6])**, and presented at ***IEEE Globecom 2025*** and ***IEEE ESSERC 2024*** with oral presentations **(refer to Conference papers [1] and [7])**.

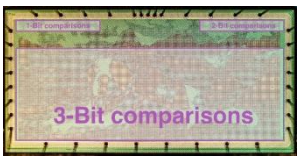
CHIP GALLERY



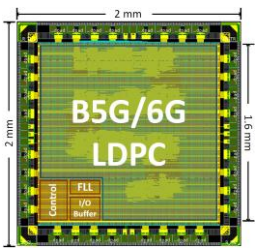
28nm NR-Polar decoder
Frame-interleaving,
3.34Gbps, 1.28mm2
all 5G polar support



28nm NR-LDPC decoder
Block-parallel
24.4Gbps, 1.83mm2
all 5G LDPC support



65nm GRAND ASIC
Unified decoding
8.84 worst-case T/P
all small matrix support



16nm B5G/6G LDPC ASIC
Multi-engine+multi-core
126 Gbps, 2.56mm2
First B5G/6G tape-out